

Revision Notes for Class 7 Science Chapter 6 Adolescence: A Stage of Growth and Change

Adolescence is a crucial phase in human development, marking the transition from childhood to adulthood. It is typically experienced between the ages of 10 and 19 years. Unlike infancy and early childhood, which are primarily marked by gradual increases in height and weight, adolescence brings rapid physical, emotional, and social changes. During this stage, the body undergoes noticeable transformations such as increasing height, weight gain, development of secondary sexual characteristics, and changes in behavior and emotions.

Stages of Human Life

Human life can be broadly divided into infancy, childhood, adolescence, adulthood, and old age. Each person passes through these stages, but the duration and experience vary from one individual to another. The shift into adolescence starts around age 10 and continues until 19. This is when most changes—both visible and invisible—take place, helping the body to prepare for adulthood.

Observable Changes During Adolescence

Some of the most common changes adolescents notice in themselves and their peers include an increase in height, weight, and physical strength. Boys tend to develop broader shoulders and wider chests, while girls may experience the development of breasts and body changes related to puberty. Both boys and girls grow hair in new places such as armpits and the pubic region; boys also start to grow facial hair like a moustache or beard.

Most Common Changes During Adolescence

S.No	Changes
1.	Increase in height

2.	Weight gain and enhanced strength
3.	Body appearance and new body features
4.	Voice changes (especially in boys)
5.	Development of pimples (acne)
6.	Growth of hair in new regions

Voice changes occur as the voice box (larynx) grows, especially in boys, causing voices to become deeper and sometimes hoarse. In boys, the "Adam's apple" appears prominently. Another visible change is the onset of acne or pimples on the face. Increased oil production clogs pores during adolescence and may lead to pimples.

Secondary Sexual Characteristics and Puberty

Adolescence introduces secondary sexual characteristics—features that distinguish males and females but are not directly involved in reproduction. Examples include breast development in girls and facial/chest hair in boys. These changes signal the arrival of puberty, a key developmental stage that enables the body to become capable of reproduction. The timing and nature of these changes vary greatly among individuals and are entirely normal.

Internal Changes: Reproductive Capability

Apart from visible signs, internal developments occur as well. For girls, puberty marks the beginning of the menstrual cycle, which typically occurs every 28–30 days but can range from 21–35 days. Menstruation usually lasts three to seven days and is a healthy, natural process that indicates reproductive health. Menstruation ceases by the age of 45–55, ending a woman's reproductive phase. Myths and taboos about menstruation are not scientifically valid and should be replaced by accurate knowledge and healthy practices.

Emotional and Behavioural Changes

Emotions tend to become stronger and more unpredictable during adolescence. Teenagers may experience mood swings, increased sensitivity, and heightened curiosity and excitement. These emotional changes might cause moodiness, self-exploration, or even involvement in creative, social, or sports activities. Understanding and managing these emotions responsibly is important for positive personal growth.

Emotional Changes During Adolescence

Key Emotional Change	Possible Behavioural Effects	Ways for Positive Growth
Mood swings	Strong emotions	Music, sports, dance, and arts
Increased sensitivity	Self-discovery, creativity	Encourage creative activities
New perspectives	Interest in community/social work	Volunteer, collaborate with peers

Tips for Healthy Adolescence

It is important for adolescents to adopt good habits during this period to ensure overall well-being. This includes eating a nutritious, balanced diet rich in proteins, carbohydrates, fats, vitamins, and minerals. Foods like milk, cereals, green leafy vegetables, and fruits supply essential nutrients such as calcium and iron, which are vital for bone growth and blood formation—especially important for girls to prevent anemia.

Sample Food Sources and Nutrients

Food Source	Nutrients	Function
Milk, curd, cheese, paneer	Calcium, proteins, fats	Bone growth
Leafy greens, beans, dried fruits	Iron	Blood formation

Maintaining personal hygiene is also essential during adolescence. Washing the body regularly, keeping the armpits and pubic region clean, and for girls, following menstrual hygiene practices (using sanitary pads or clean cloth pads and disposing them responsibly) are crucial steps to avoid infections and stay healthy.

Physical Activities and Social Well-being

Regular exercise, sports, and other physical activities help adolescents maintain physical fitness, build stamina, and improve concentration. Interacting respectfully with peers, both face-to-face and online, is necessary for a safe and supportive social environment. Adolescents must remember to use social media responsibly and guard against cyberbullying.

- Eat healthy and balanced meals every day.
- Practice hygiene, especially during menstruation.
- Stay physically active and exercise regularly.
- Interact politely in society and on digital platforms.
- Seek guidance from elders when needed.

Avoiding Harmful Substances

Adolescents should completely stay away from harmful substances like tobacco, alcohol, drugs, and similar addictive items. These not only damage health but may also lead to addiction, affecting future prospects. If pressured or tempted, it is important to be firm and say “NO.” Those struggling with substance abuse should seek help from family, friends, teachers, or helplines such as the National De-addiction Helpline (14446).

Why Do These Changes Happen?

The changes experienced during adolescence are primarily controlled by hormones—special chemicals produced by various glands in the body. These hormones regulate growth, emotions, and the development of secondary sexual characteristics.

Hormones are released in response to signals from the brain and function at specific times to prepare the adolescent body for adulthood.

Key Points in a Nutshell

- Adolescence occurs between 10–19 years and is marked by major physical and emotional changes.
- Secondary sexual characteristics appear—breast growth in girls, facial hair in boys.
- Puberty signifies the beginning of reproductive capability.
- Girls begin menstruation, which generally lasts till 45–55 years of age.
- Nutrition, hygiene, exercise, and avoiding addiction are crucial for a healthy adolescence.
- Emotional changes can be managed positively with awareness and support.
- All adolescent changes are mainly controlled by hormones.

By understanding and accepting these changes, and by making informed and healthy choices, adolescents can enjoy this exciting and vital stage of life with confidence and responsibility.